# Bimlesh Singh

Data Science & Analytics Lead
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## Profile Summary

Data Science & Analytics Leader with 12+ years of experience delivering business impact across BFSI, Retail & CPG, and Manufacturing domains. Proven expertise in financial crime analytics, credit risk, fraud detection, and customer analytics, alongside leadership in building cross-functional data teams, delivering GenAI and driving AI/ML solutions that optimize pricing, demand forecasting, promotions. Skilled in translating business problems into scalable data solutions.

## Educations

**MA – Applied Statistics** 2017
IGNOU |New Delhi, India

**B. Tech** 2011
Amity University| Noida, India

## Experience

**Data Science Architect** Feb 2024 - Present
Nagarro Remote

- Collaborated with senior leaders across industries to shape analytics strategy, enable data-driven decision making.
- Led a team and mentored of 7+ data scientists, analysts delivering data science and analytics solutions and implementing coaching frameworks to ensure high performance.
- Worked closely with sales and business development teams to create impactful POCs and proposals for clients; successfully converted multiple POCs into long-term engagements and projects.
- led the team to designed and implemented AI-driven pricing models for building materials by analysing the impact of price changes on customer demand, resulting in a 2% increase in profit margins across 50 product.
- Designed and deployed demand forecasting solutions for 500+ SKUs for manufacturing client, reducing forecast errors by 11% and improving supply chain planning.
- Presented forecasting insights to CXO-level stakeholders, influencing strategic planning around battery storage deployment, renewable energy investments, and supply chain readiness.
- Developed forecasting models for battery energy equipment sales in the Germany market, incorporating drivers such as energy consumption, market capacity, and macroeconomic factors.
- Led a cross-functional team of GenAI and ML engineers to design and deploy a Chabot for stakeholders, using OpenAI API and LangChain; ensured seamless UI integration and alignment with relationship management workflows, reducing manual effort by 20%.
- Directed team to developed scalable GenAI pipelines integrating OpenAI GPT4.o with internal knowledge bases using RAG architecture, enabling finance users to retrieve contextual answers from complex policy and financial documents.
- Directed a data science initiative to design cross-sell recommendation systems, delivering a 10% uplift in product sales and enabling sales teams to target high value customers.

**Principle Consultant** Sep 2023 -Feb 2024
Quara AI Remote

- Established the data science function from the ground up , including team structure, recruitment, and capability building. Mentored and coached junior data scientists, driving knowledge transfer.
- Partnered with product & engineering teams to introduce an “investment growth return” feature on property pages, enhancing user experience and reducing bounce rate by 20%.

**Lead Data Scientist** April 2022- Jul 2023
IQVIA Bangalore, India

- Partnered with product & engineering managed collaborated closely with cross-functional teams, including marketing, pharma etc., to identify business challenges and develop solutions using analytics and machine learning.
- Development and deployment of predictive models to forecast patient risk reducing hospital readmission rates by 9%.
- Led a cross-functional team of data scientists and analysts to design and implement a Market Mix Model for pharma business, enabling data-driven allocation of marketing budgets and delivering improvement in ROI
- Implemented claims fraud detection models for healthcare provider reducing fraudulent activities by 10%.
- Led the design and deployment of Tableau dashboards for a health insurance client, providing real-time visibility into claims, policy KPIs; enabled executives to make faster data-driven decisions and reduced manual reporting effort by 40%.

**Data Science Manager , Financial Crime Analytics** Mar 2020 – Apr 2022
CITI Pune, India

- Built graph-based ML models for network analysis, identifying hidden relationships in money laundering schemes.
- Led the design and deployment of ML-driven shell company detection models, increasing identification rate by 5% and enhancing AML program effectiveness; collaborated with compliance and risk teams to investigate further.
- **Designed dashboards and reporting solutions** (SAS, Tableau, SQL) to monitor AML key risk indicators (KRIs) and provide transparency to risk management teams

**Data Scientist** July 2018- Mar 2020
R Systems International Ltd Noida

- **Developed customer segmentation frameworks** using spend patterns, demographics, and utilization data, enabling personalized offers and improving cross-sell conversion by **12%**.
- Built and delivered campaign dashboards and performance trackers in SQL and Tableau, providing real-time insights to marketing teams.
- **Implemented cross-sell and upsell propensity models**, identifying high-value customers and driving **incremental revenue uplift** in the card portfolio.
- **Led the development of a computer vision solution leveraging transfer learning and CNNs with fine-tuning**, achieving **80% accuracy** in table detection and extraction from images, and enabling automation of reporting workflows.

**Analytics Manager** April 2015- Dec 2017
Apollo Munich Health Insurance Gurgaon

- Designed dashboards and KPI trackers that empowered senior management to monitor performance trends and pivot strategies for decision making.
- Guided team in building **churn models** at the time of renewal that improved retention by **3%**.
- Created a data-driven forecasting model to predict disease outbreaks, enabling the company to better anticipate claims demand and implement targeted policies for effective claims management.
- Utilized historical patterns to segment customers based on health consciousness, enabling the sales team to make more informed decisions and improve selection risks.

**Campaign Analyst, Credit Card** Aug 2014- April 2015
HSBC Bangalore

- Campaign Performance Analytics: Analysed post-campaign data to assess performance, generating actionable insights that drove increase in engagement and response rates.
- Leveraged SQL and SAS for advanced data analysis, working with large datasets to identify actionable insights.

**Business Analyst** Nov 2012- Aug 2014
Times Internet Ltd Gurgaon

- Marketing KPI Tracking & Reporting: Designed and maintained dashboards in Tableau, providing real-time insights into key marketing metrics

## Core Skills

- **Machine Learning & AI:** Linear & Logistic Regression, Decision Trees, Random Forest, XGBoost, Boosting, KNN, SVM, K-Means, DBSCAN, Ridge & Lasso, Market Basket Analysis, Deep Learning (LSTM, RNN, CNN, Transformers), Anomaly detection.
- **Optimization & Forecasting:** Linear Programming, Demand Forecasting, Price Elasticity, Price Optimization, ARIMA/SARIMA/SARIMAX.
- **Analytics Expertise:** Segmentation, Predictive Modelling, Applied Statistics, Hypothesis Testing, A/B Testing, Customer Analytics, Marketing Analytics, CLTV & Churn Modelling, Customer Life Cycle.
  Cross & Upsell.
- **Leadership & Strategy:** Team Building & Development, AI/ML Strategy, Executive Advisory, Business-Aligned Analytics, Stakeholder Engagement, Team Mentorship.

## Knowledge & Exposure

- **Domain :** Credit Risk Analytics, Model validation, Credit Card Analytics, Underwriting.

## Tools

- **Programming & Platforms:** Python, SQL, SAS, Azure Machine Learning, Databricks, SAS E-Miner, SPSS
- **Visualization & BI:** Tableau, Power BI, MS Access.
- **Generative AI:** Large Language Models (LLMs), LangChain, GPT-4, Retrieval-Augmented Generation (RAG), AI Agents, Vector Database, Tool Integration.
- **MLOps & Deployment:** Azure ML, Databricks, Docker, Fast API, Flask, Model Deployment Pipelines
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